Proposal for Measurement Method of Transfer Impedance of Current Probe

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***Abstract*—This paper proposes a measurement method with a high reliability and reproducibility for the transfer impedance of current probe. The measurement method uses a vector network analyzer and a calibration fixture. The advantages of the proposed measurement method are the high reliability and reproducibility of measured results among different measurement systems, and the reduction of measurement error of the used measurement system. From the experimental measurements, it is shown that the proposed measurement method is more effective than the conventional methods.**

***Index Terms*—Calibration fixture, current probe, measurement method, reliability, reproducibility, transfer impedance, vector network analyzer (VNA).**

I. Introduction

THE measurement method for transfer impedance of current probe has been provided by international standards such as Comité International Spécial des Perturbations Radioélectriques (CISPR) or International Organization for Standardization (ISO) [1], [2]. Also, most calibration laboratories and current probe manufacturers might be applying the original measurement method in compliance with these standards. However these conventional measurement methods seem to have some problematic issues.

In CISPR 16-1-2, the measurement method uses an appropriate coaxial structure jig under the well-matching condition of the coaxial line [1]. The first issue is that most current probe manufacturers do not provide the coaxial structure jig. The second issue is that the standard does not regulate the matching degree of the coaxial line.

Also, in ISO 11452-4, the measurement method uses a vector network analyzer (VNA) and a calibration fixture (or it might be called a calibration jig) [2]. The calibration fixture is regulated to have voltage standing wave ratio (VSWR) <1.2. But the standard does not specify the calibration method of the measurement system including the VNA and the calibration fixture.

Moreover, the FCC, Inc. (FISCHER CUSTOM COMMUNICATIONS, INC.), which is one of the current probe manufacturers, provides the original measurement method in compliance with ISO 11452-4 [3], [4]. However it does not also give enough information for the calibration of the VNA.

These conventional methods do not use the same measurement system configuration for the transfer impedance measurement and the same calculation formula for the transfer impedance from measurement data. So the transfer impedance measurement data obtained by different methods using different measurement system configurations will not be the same. Therefore, the measurement data providing the transfer impedance characteristics of the current probe might lack the reliability.

This study has been conducted to develop a measurement method with high reliability transfer impedance measurement result, which is available in the use of different equipment and cables on the same measurement system configuration. The highly reliable measurement method has to have a high reproducibility of measured result. This paper proposes a transfer impedance measurement method with a high reliability and reproducibility, by means of the VNA and the calibration fixture.

In this paper, the aforementioned conventional measurement methods are first introduced and the problematic issues are pointed out. Next, the measurement method of transfer impedance by means of the VNA and the calibration fixture is proposed. The measurement principle is simple, understandable, and more strict than the conventional methods. The validity of the proposed measurement method was verified by comparison with the conventional measurement method of FCC, Inc. in the experimental measurements. Then, the calibration method of the VNA was also investigated. Furthermore, the influence of the current probe installed position in the calibration fixture on the transfer impedance measurement results is discussed. As a consequence, the maximum measurement error relative to the averaged measurement results was experimentally confirmed to be less than 1%.

The proposed measurement method can provide the transfer impedance measurement results with a high reliability and reproducibility. In addition, it was confirmed that the proposed measurement method was available for the different VNAs and test cables.

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II. Conventional Measurement Methods and Their Problematic Issues

The principle operation of current probe is same as the current transformer, which detects the time varying current on the conducting wire or line in the circuit [5]. This type of current probe is called ac (alternating current) current probe. The time varying current measured on the line is called primary current. Also, the current probe is usually connected to a measuring instrument

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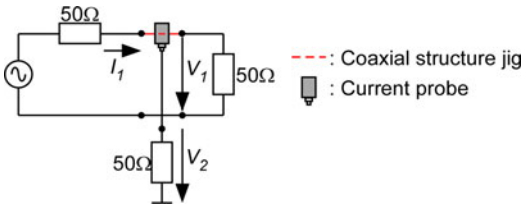


Fig. 1. Schematic diagram of equivalent calibration circuit for measuring transfer impedance of current probe provided by CISPR 16-1-2.

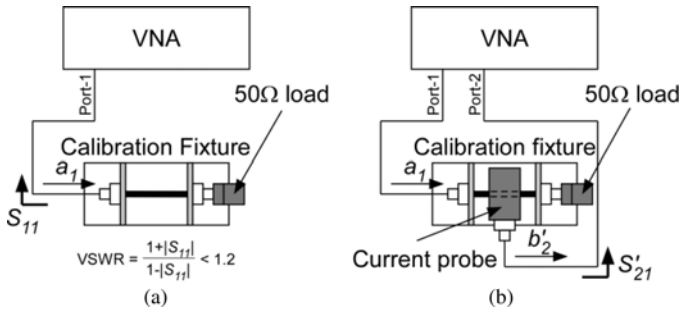


Fig. 2. Measurement configurations for the calibration fixture and the current probe. (a) For fixture calibration. (b) For current probe.

with internal resistance of 50 Ω. Thus, the current probe output is readable as voltage. It is called secondary voltage. The current detection sensitivity is presented by the conversion ratio of the secondary voltage to the primary current, i.e. Ω or dBΩ. It is generally called transfer impedance. So the transfer impedance is one of the crucial characteristics of the current probe. It differs with the frequency of the measuring current.

There are some interesting research studies concerning the transfer impedance measurement of current probe. Ruddle *et al.* proposed a simple 1-D transmission line model to derive the transfer impedance of current transducers at high frequencies, which used additional measurements to quantify the mismatches in the system [6]. Pommerenke *et al.* proposed a method with a network analyzer for measuring directly the current at the center of the current clamp, which optimizes the test setup to a point at which no data correction is needed anymore [7].

The purpose of this research is development of the transfer impedance measurement method with a high reliability and reproducibility without depending on the used measurement system. This section introduces some different measurement methods in use and their problematic issues.

A. CISPR 16-1-2

CISPR 16-1-2 provides the measurement method for the transfer impedance of current probe as following [1]. Fig. 1 shows the schematic diagram of the equivalent calibration circuit for measuring the transfer impedance of current probe. The dashed red line in Fig. 1 presents the measurement jig, which is made of two halves of a coaxial adaptor. The current I_1 through the inner conductor in the coaxial structure jig is calculated from the voltage V_1 on the primary line with 50 Ω load, when the coaxial line is well matched. Also, the current probe detects the current I_1 as the secondary voltage V_2. Thus, the transfer impedance Z_T Ω can be calculated from the following equation:

$$Z_T = \frac{V_2}{I_1} = 50_{[\Omega]} \cdot \frac{V_2}{V_1}. \tag{1}$$

The measurement principle is simple, understandable, and correct. But the biggest problematic issue is that most manufacturers today do not provide the coaxial structure jig. Moreover, the standard does not regulate the impedance matching degree among the jig, the measurement equipments, and the lines or cables. Thus, the transfer impedance measurements give different results depending on the impedance matching degree.

B. ISO 11452-4

In ISO 11452-4, the measurement method for the transfer impedance of current probe is provided as following [2]. The measurement configurations for the calibration fixture and the current probe are shown in Fig. 2(a) and (b), respectively. The calibration fixture in Fig. 2(a) is connected to the Port-1 of the VNA and 50 Ω load for the calibration. And this standard gives VSWR <1.2 for the calibration fixture in the measurement configuration. So the most part of the input signal a_1 from the Port-1 is transmitted to the 50 Ω load. In brief, the measurement configuration in Fig. 2(a) is well-matching, but not perfect-matching. The current probe in Fig. 2(b) is clamped to the inner conductor of the calibration fixture, and its output terminal is connected to the Port-2 of the VNA. The Port-2 measures the signal b'_2 detected by the current probe as the transmission coefficient S'_{21}. The transfer impedance Z_T in (1) can be also expressed by the power of each port. So the magnitude of the transfer impedance Z_T is calculated directly from the measured transmission coefficient S'_{21} as follows:

$$|Z_T| = 50_{[\Omega]} \cdot |S'_{21}|. \tag{2}$$

This measurement method has three problematic issues. The first issue is that the magnitude of the calculated transfer impedance Z_T has a small difference in the use of different calibration fixtures satisfying the VSWR <1.2. The second issue is that the VSWR of the calibration fixture changes with the current probe inserted into the calibration fixture, because the input signal a_1 from the Port-1 reflects slightly on the coupling impedance of the inner conductor of the calibration fixture with the current probe installed. Namely, the current I_{a1} toward the 50 Ω load is different before and after inserting the current probe into the calibration fixture. As a consequence, (2) may not be appropriate for the accuracy of the transfer impedance measurement result. Essentially, the calibration in Fig. 2(a) should take into account the current probe installed in the calibration fixture. The last issue is that the standard does not specify the calibration method of the VNA and the measurement parameters such as the intermediate frequency (IF) bandwidth and the output power, even though they affect the objective measurement data [8], [9]. Therefore, the transfer impedance measurements with the different calibration methods or measurement parameters do not give the same result.

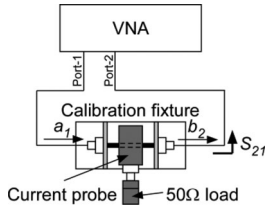


Fig. 3. Calibration configuration for VNA with dedicated calibration fixture.

C. Current Probe Manufacturer

The FCC, Inc. provides the original measurement method in compliance with ISO 11452-4 [3], [4]. Fig. 3 shows the calibration configuration of the VNA with the dedicated calibration fixture [3]. The VNA is first calibrated by the thru response calibration by means of the dedicated calibration fixture with the tested current probe and 50 Ω load installed. This calibration eliminates the frequency response transmission tracking error for the test setup [8], [9]. After performing the calibration, the transmission coefficient S_{21} through the current probe is measured by the measurement system configuration as on Fig. 2(b). The magnitude of the transfer impedance Z_T of the current probe is then calculated by (2).

The measurement method is compliant to ISO 11452-4 except for the thru response calibration of the VNA. However, it has some problematic issues. The one is that the measurement method does not provide any electrical parameters of the calibration fixture such as offset delay, offset loss, and offset characteristic impedance, which are necessary for the thru response calibration. The other is that the thru response calibration cannot correct essentially the impedance mismatch in the measurement system including the VNA and the test cables [8], [9]. Thus, the transfer impedance measurement gives hovering results (see Section IV).

III. Proposed Measurement Method

In this section, a transfer impedance measurement method with a high reliability and reproducibility is proposed by means of the VNA and the calibration fixture that is a noninsertable device. The calibration of the VNA is described later.

First, the proposed measurement method uses the measurement system configuration as shown in Fig. 3. Then, the Port-2 of the VNA measures the signal b_2 through the calibration fixture with the installed current probe as the transmission coefficient S_{21} of the calibration fixture. The measured transmission coefficient S_{21} is used to remove the influence of the input signal reflection stemming from the current probe inserted into the calibration fixture.

Second, the measurement system configuration as shown in Fig. 2(b) is used. Then, the Port-2 of the VNA measures the signal b_2' detected by the current probe as the transmission coefficient S_{21}' of the current probe. Note that the input signal a_1 into the calibration fixture in Figs. 3 and 2(b) is almost the same, because the 50 Ω load connected to the calibration fixture or to the current probe is almost equal to an internal resistance of the VNA.

TABLE I
Measurement Accuracy and Usable Calibration Methods for Respective Models of VNA Provided by Agilent Technologies, Inc.

Calibration method / Model number	Thru Response	Defined Thru	Adapter Removal
E5061A	Poor	Good	Unacceptable
E5061B	Poor	Good	Excellent
E5071C	Poor	Good	Excellent

Finally, the magnitude of the transfer impedance Z_T can be calculated as the ratio of two transmission coefficients S_{21} and S_{21}' as follows:

$$Z_T = \frac{V_2}{I_1} = 50_{[\Omega]} \cdot \frac{V_2}{V_1} = 50_{[\Omega]} \cdot \frac{S_{21}'}{S_{21}}. \tag{3}$$

Note that the transfer impedance Z_T is represented as the ratio of the secondary voltage V_2 detected by the current probe to the primary current I_1 into the calibration fixture as in (1). So (3) can be derived from (1), because of the 50 Ω system using the VNA. The measurement principle is simple, understandable, and more strict than the conventional methods. So the transfer impedance measurement results become also more accurate.

The VNA with four ports can connect the calibration fixture to the Port-2 and the current probe to the Port-3 in Fig. 3. So the VNA can measure two transmission coefficients of S_{21} and S_{31} at a time. The transfer impedance Z_T can be then calculated in highest accuracy from the S_{21} and S_{31} transmission coefficients.

Note that the VNA can perform the thru response calibration, the defined thru calibration, the adapter removal calibration and so on for the noninsertable device [8], [9]. The used calibration method depends on the model of VNA and the measurement accuracy. As an example, Table I shows the measurement accuracies for the calibration methods and the models of VNA provided by Agilent Technologies, Inc. From Table I, the adapter removal calibration gives the highest measurement accuracy for the E5061B and E5071C models.

IV. Comparisons With Measurement Method by FCC, Inc. and Proposed Measurement Method

The transfer impedance of a current probe was experimentally measured by the conventional measurement method of Section II-C and by the proposed measurement method of Section III. The tested current probe and the dedicated calibration fixture were FCC F-61 and FCC-MPCF-3-32/71/19 manufactured by FCC, Inc. [3], [4]. Fig. 4(a) and (b) shows the exteriors of the current probe and the dedicated calibration fixture, respectively. Table II shows the main specifications of the current probe. Fig. 5 shows the frequency characteristics of the VSWR of the dedicated calibration fixture, and it does not satisfy the ISO 11452-4 regulation of VSWR $<$1.2 in the frequency region $>$300 MHz.

In experimental measurements, the used VNAs were the E5061B and E5071C models from Table I. The measurement frequency was up to 1000 MHz taking into account the usable maximum frequency of the current probe. The IF bandwidth and the output power of the VNA were 100 Hz and 0 dBm,

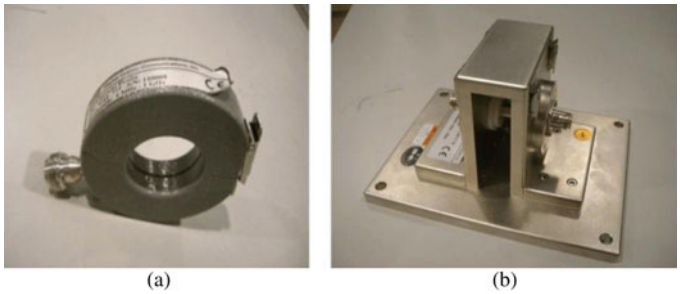


Fig. 4. Exteriors of current probe FCC F-61 and calibration fixture FCC-MPCF-3-32/71/19. (a) Current probe. (b) Calibration fixture.

TABLE II
MAIN SPECIFICATIONS OF CURRENT PROBE FCC F-61

Usable frequency	30MHz - 1GHz (-3dB bandwidth)
Internal diameter	32mm
External diameter	71mm
Height	19mm
Transfer impedance	16Ω (24dBΩ)
Connector	Type N

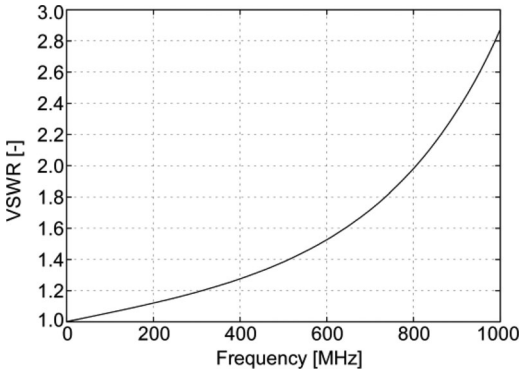


Fig. 5. Frequency characteristics of VSWR for calibration fixture FCC-MPCF-3-32/71/19.

TABLE III
SUMMARIZED SPECIFICATIONS OF USED SHIELDED ROOM

Size	Length 5m x Width 4m x Height 3m
Shield properties	Electric field: >100dB (10kHz - 30MHz) Plane wave: >100dB (30MHz - 6GHz) (IEEE-STD-299 and MIL-STD-285 Compliant)
Power filter	>100dB (14kHz - 6GHz)

respectively, which were referred from the data sheet of the current probe [4]. Incidentally, the experimental measurements were then conducted in a shielded room in order to remove the influence of external noise. Table III shows the summarized specifications of the shield room manufactured by the Riken Environmental System Co., Ltd.

A. Conventional Measurement Method by FCC, Inc.

The thru response calibration for the E5061B and E5071C models was performed in the measurement system configuration as shown in Fig. 3. The calibration did not set the electrical parameters of the calibration fixture such as offset delay, offset loss, and offset characteristic impedance, because these parameters were not provided [3], [4]. Namely, they were set to zero. After calibrating the E5061B and E5071C models, the transmission coefficient S'_{21} through the current probe was measured in the measurement system configuration as shown in Fig. 2(b). The magnitude of the transfer impedance Z_T was then calculated by (2).

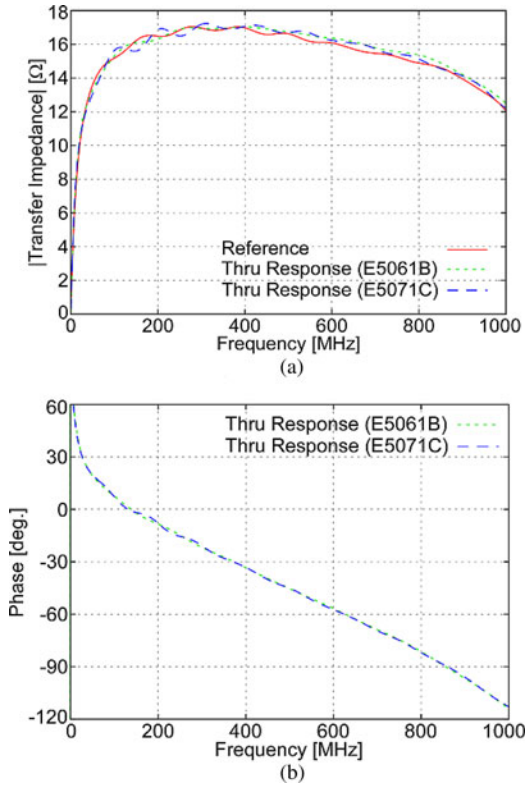


Fig. 6. Measurement results of transfer impedance of current probe FCC F-61 by E5061B and E5071C using thru response calibration. (a) Magnitude. (b) Phase.

Fig. 6(a) and (b) shows the measured results of the magnitudes and the phases of the transfer impedance Z_T, respectively. The solid red line in Fig. 6(a) shows the reference result of the E5071C model provided by the FCC, Inc. The dotted green line and the dashed blue line in Fig. 6(a) and (b) show the measured results of the E5061B and E5071C models, respectively. The magnitudes and the phases of all results in Fig. 6 are almost the same degree, but these results do not completely coincide. These frequency characteristics show nonmonotonous variation with different fluctuation. The fluctuation is caused by multiple reflections of signal stemming from an impedance mismatch degree in the measurement system including the VNA, the test cables and so on [8], [9]. Note that both the solid red line (the reference result) and the dashed blue line (the measured result) in Fig. 6(a) are obtained from the same VNA (E5071C), but the fluctuation of them does not coincide. It is considered that the difference in the fluctuation is attributed to the used test cables and coaxial conversion connectors. As a consequence, it was found that the conventional measurement method by FCC, Inc. gives the fluctuation of the transfer impedance measurement results depending on the measurement system. The fluctuation is

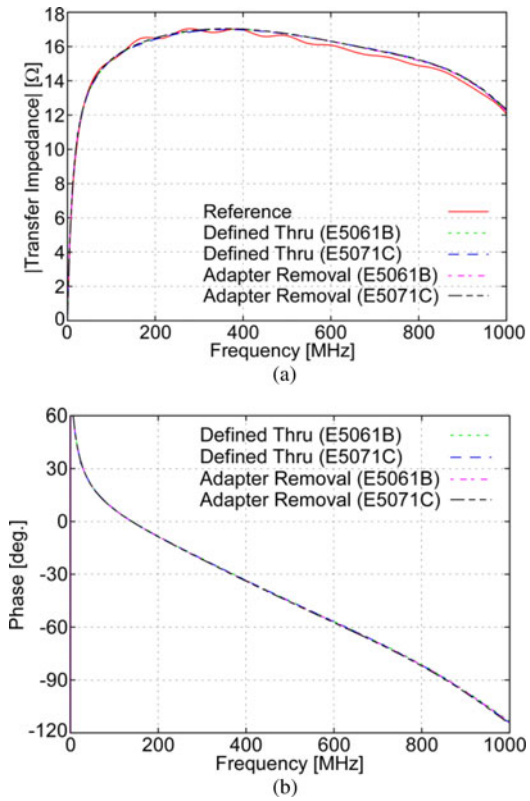


Fig. 7. Measurement results of transfer impedance of current probe FCC F-61 by E5061B and E5071C using thru response and adapter removal calibrations. (a) Magnitude. (b) Phase.

caused by the impedance mismatch in the measurement system uncorrected by the thru response calibration.

B. Proposed Measurement Method

The proposed measurement method was performed using the E5061B and E5071C models. From Table II, the calibration methods of the VNAs selected the defined thru and adapter removal calibrations. After calibrating the VNAs by each calibration method, the magnitudes and the phases of the transfer impedance Z_T were measured according to the proposed measurement method in Section III.

The magnitudes and the phases of the transfer impedance measurement results are shown in Fig. 7(a) and (b), respectively. The solid red line in Fig. 7(a) shows the reference result of the E5071C model provided by the FCC, Inc., as well as in the previous section. In Fig. 7(a) and (b), the dotted green line, the dashed blue line, the dashed–dotted pink line, and the double-dashed black line show the measured results of the E5061B and E5071C models calibrated by the defined thru calibration and the adapter removal calibration, respectively.

In Fig. 7(a), the magnitude of the reference result of the FCC, Inc. shows nonmonotonous variation with fluctuation, as mentioned in the previous section. In contrast, the measured results show monotonous frequency characteristics without fluctuation. In the frequency region <400 MHz, the measured results resides in the fluctuation of the reference result. In the frequency region from 400 to 1000 MHz, the measured results become larger

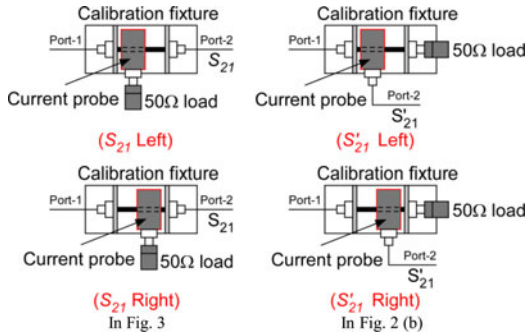


Fig. 8. Install positions of current probe in calibration fixture in Figs. 3 and 2(b).

than the reference result. The maximum difference is about 0.40 Ω. In addition, the differences among the measured results are smaller than 0.15 Ω, and the maximum measurement error is less than about 0.7% relative to the averaged measurement result. In Fig. 7(b), the measured results almost coincide without fluctuation. So, it was found that the defined thru and adapter removal calibration methods correct well the impedance mismatch in the measurement system including the VNA and the test cables.

The adapter removal calibration is essentially more accurate than the defined thru calibration, but the measured results hardly differ. It is considered that the reason of this was the measurement frequency, which was up to 1000 MHz. This aspect will depend on the relation between the physical length of the calibration fixture and the measurement wavelength [8], [9].

Incidentally, the similar experimental measurements were performed for the different current probe FCC F-65A using the same dedicated calibration fixture. The maximum measurement error was less than 1.0% relative to the averaged measurement result in the frequency range <1000 MHz. The other results were similar to the aforementioned.

V. Discussion on Position of Current Probe Installed in Calibration Fixture

The used calibration fixture has the spacer on both sides of the inner conductor to install the tested current probe. The installed position might move sideways by a few millimeters. Therefore, the influence of installed position on the transfer impedance measurement result was investigated by the proposed measurement method. This investigation is important in developing the transfer impedance measurement method with a high reliability and reproducibility.

Fig. 8 shows the positions of the current probe on the calibration fixture in Figs. 3 and 2(b). (S_{21} Left), (S_{21} Right), (S'_{21} Left), and (S'_{21} Right) represent the position of current probe at the left side and right side in the calibration fixture in Figs. 3 and 2(b), respectively. Now, the transmission coefficient S_{21} through the calibration fixture and the transmission coefficient S'_{21} through the current probe were measured at each position. The measurement conditions are the same as in Section IV.

Fig. 9(a) and (b) shows the magnitudes and the phases of the transfer impedance measurement results of the E5071C model

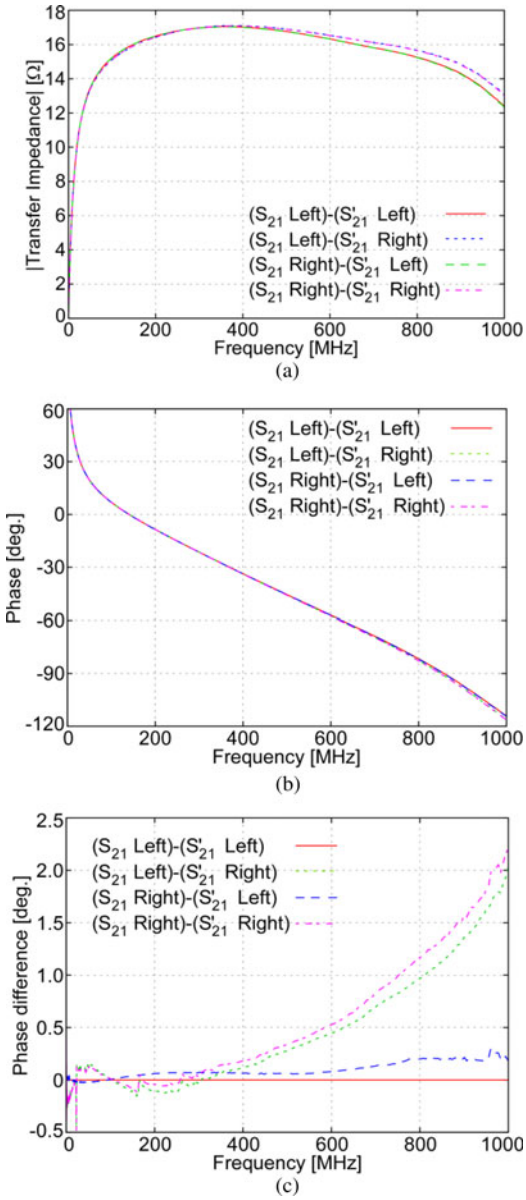


Fig. 9. Measurement results of transfer impedance of current probe FCC F-61 by E5071C using adapter removal calibration, when changing install position of current probe in calibration fixture. (a) Magnitude. (b) Phase. (c) Each phase difference based on (S21 Left)–(S′21 Left).

calibrated by the adapter removal calibration. The solid red line, the dotted green line, the dashed blue line, and the dashed–dotted pink line are the measured results of the following combinations: (S_{21} Left)–(S'_{21} Left), (S_{21} Left)–(S'_{21} Right), (S_{21} Right)–(S'_{21} Left), and (S_{21} Right)–(S'_{21} Right), respectively. Additionally, Fig. 9(c) shows each phase difference of (S_{21} Left)–(S'_{21} Right), (S_{21} Right)–(S'_{21} Left), and (S_{21} Right)–(S'_{21} Right) based on (S_{21} Left)–(S'_{21} Left).

From Fig. 9(a) and (b), one can find that the measured results were affected by the combinations of the position of the current probe. The difference among the phases is clearly seen from Fig. 9(c). Comparing with the combinations in details, the installed position affected not the transmission coefficient S_{21} through the calibration fixture, but the transmission coefficient S'_{21} through the current probe. The influence became larger at the higher measurement frequency. The maximum difference among the magnitudes in Fig. 9(a) and the phases in Fig. 9(c) is about 0.7 Ω and 2.2° at 1000 MHz, respectively. The difference 0.7 Ω among the magnitudes may be stemmed from the difference 2.2° among the phases. Incidentally, the differences of the phases at the same installed positions in measuring the transmission coefficient S'_{21} through the current probe (namely, the solid red line and the dashed blue line, and the dotted green line and the dashed–dotted pink line) are about 0.3°. This slight difference may be caused by the human setting error of the current probe position. The detailed aspects about the relation between the installed position and the phase of the transfer impedance measurement result are under consideration.

As a consequence, the position of the current probe installed in the calibration fixture should be specified exactly to get a high reliability and reproducibility of the transfer impedance measurement.

VI. Conclusion

This paper proposes the measurement method with a high reliability and reproducibility for the transfer impedance of a current probe by means of a VNA and a calibration fixture. The proposed measurement method is as follows.

1) The VNA with test cables is calibrated with the most accurate method in the usable full two port calibrations. Note that the calibration method gives high accuracy and reliability of the measurement data regardless of the models of VNA and the test cables. In this paper, the defined thru calibration and the adapter removal calibration were applied on the E5061B and E5071C models of Agilent Technologies, Inc. for the noninsertable calibration fixture.
2) The transfer impedance is calculated using (3) from the ratio of the transmission coefficients of the calibration fixture in Fig. 3 and the current probe in Fig. 2(b). Note that the measurement principle is simple, understandable, and more strict than the conventional methods. The measurement method of the transmission coefficients can remove the influence of the input signal reflection stemming from the current probe installed in the calibration fixture.
3) To get higher reliability and reproducibility for the transfer impedance measurement result, the position of current probe installed in calibration fixture should be taken into consideration. If possible, the position should be specified exactly.

As a consequence, the proposed measurement method can give higher reliability, reproducibility, and accuracy than the conventional methods for the transfer impedance measurement of current probe. In the experimental measurements by the proposed measurement method, it was confirmed that the maximum measurement error was less than 1.0% relative to the averaged measurement result. It is more effective for the reliability and reproducibility of the magnitude of the transfer impedance in the frequency region <400 MHz especially. Furthermore, one of the advantages is that the measured result does not depend on

the used measurement system including the VNA and the test cables.

In future works, the uncertainty budget for each measurement method will be estimated and compared to get the theoretical verification of the proposed measurement method [10].

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